40.520 Stochastic Models

Poisson Processes

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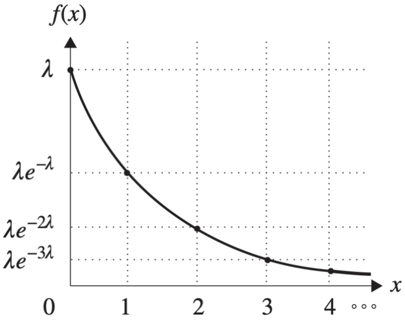
Outline

The Exponential Distribution

Definition

- A continuous random variable X is said to have an **exponential distribution** with parameter λ: $X \sim Exp(\lambda)$ if its probability density function is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0; \\ 0, & x < 0. \end{cases}$$



CDF, Mean and Variance

- The CDF of an exponentially distributed random variable is given by

$$F(x) = \int_{-\infty}^{x} f(u)du = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0; \\ 0, & x < 0. \end{cases}$$

- The mean and variance of an exponentially distributed random variable are
  - Mean: $E[X] = \frac{1}{\lambda}$
  - Variance: $Var[X] = \frac{1}{\lambda^2}$

The Memory-Less Property

- We can show that
$$P(X > s + t \mid X > t) = P(X > s)$$
for a random variable with exponential distribution.
- If we think of X as being the lifetime of some instrument, then this property states that the instrument does not remember that it has already been in use for a time t.

The Memory-Less Property

- The only continuous distribution to have this property is the exponential distribution.

Example: Service Time in a Bank

- Suppose that the amount of time one spends in a bank is exponentially distributed with mean 10 minutes, that is, $\lambda = 0.1$.
- What is the probability that a customer will spend more than 5 minutes in the bank?
- What is the probability that a customer will spend more than 15 minutes in the bank, given that she is still in the bank after 10 minutes?

Several Further Properties of Exponential Distribution

Suppose that $\{X_1, X_2, \ldots, X_n\}$ are independent exponential random variables, with X_i having rate λ_i, $i = 1, 2, \ldots, n$.

The distribution of

$$Y = \min\{X_1, X_2, \ldots, X_n\}$$

follows an exponential distribution with a rate $\sum_{i=1}^{n} \lambda_i$. That is,

$$P(Y > x) = e^{\left(-\sum_{i=1}^{n} \lambda_i\right)x}.$$

Several Further Properties of Exponential Distribution

Suppose that X_1 and X_2 are independent exponential random variables with rates λ_1 and λ_2, respectively. The probability

$$P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Example: Post Office

- Consider a post office that is run by two clerks. Suppose that when Mr. Smith enters the post office he discovers that Mr. Jones is being served by one of the clerks and Mr. Brown by the other. Suppose also that Mr. Smith is told that his service will begin as soon as either Jones or Brown leaves.
- If the amount of service time for each clerk is exponentially distributed with rate λ, what is the probability that, of the three customers, Mr. Smith is the last one to leave?

Example: Post Office Again

- Suppose that you arrive at a post office having two clerks at a moment when both are busy but there is no one else waiting in line. You will enter service when either clerk becomes free.
- If service times for clerk i are exponential with rate λ_i, $i = 1, 2$, find $E[T]$, where T is the amount of time that you spend in the post office.

Several Further Properties of Exponential Distribution

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed exponential random variables having mean $1/\lambda$. The sum

$$Z_n = X_1 + X_2 + \ldots + X_n$$

has a gamma distribution; that is, its PDF is given by

$$f_{Z_n}(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}.$$

The Poisson Process

Counting Processes

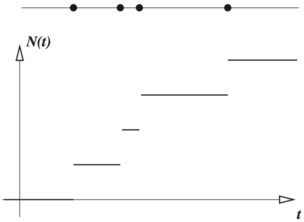
A stochastic process $\{N(t),\ t \geq 0\}$ is said to be a *counting process* if $N(t)$ represents the total number of "events" that have occurred up to time t. Hence, a counting process $N(t)$ must satisfy:

(i) $N(t) \geq 0$.

(ii) $N(t)$ is integer-valued.

(iii) If $s < t$, then $N(s) \leq N(t)$.

(iv) For $s < t$, $N(t) - N(s)$ equals the number of events that have occurred in the interval $(s, t]$.



Counting Processes

- A counting process is said to possess **independent increments** if the numbers of events that occur in disjoint time intervals are independent.
- A counting process is said to possess **stationary increments** if the distribution of the number of events that occur in any interval of time depends only on the length of the time interval.

Construction of a Poisson Process

- Begin with a sequence $\tau_1, \tau_2, \ldots$ of independent exponential random variables, all with the same mean $1/\lambda$.
- We build a model in which "events" occur from time to time. In particular,
  - the first event occurs at time τ_1;
  - the second event occurs τ_2 time units after the first event;
  - ...

Construction of a Poisson Process

- The random variables $\{\tau_k\}_{k\in\{1,2,\cdots\}}$ are called the inter-arrival times.
- The waiting time of the n-th event is

$$S_n = \sum_{k=1}^{n} \tau_k.$$

- A process $\{N_t, t \geq 0\}$ is said to be a Poisson process if it counts the number of events in the above model that occur at or before time t.

Distribution of Poisson Processes

- We have

$$P(N_t = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \quad k = 0, 1, 2, \ldots$$

i.e., N_t follows a Poisson distribution.

Mean and Variance of Poisson Increments

Mean:

$$E[N_t - N_s] = \lambda(t - s).$$

Variance:

$$Var[N_t - N_s] = \lambda(t - s).$$

Poisson Process as a Counting Process

- In summary, the Poisson process must satisfy:
  - $N_t \geq 0$;
  - N_t is integer valued;
  - $N_t \geq N_s$ for $t \geq s$;
  - $N_t - N_s$, the number of events that occur in the interval $(s, t]$, follows a Poisson distribution.
- The Poisson process is a special case of counting processes.

Poisson Process

Definition 1 of the Poisson Process

A Poisson process having rate λ is a counting process such that

1. $N(0) = 0$.
2. The process has independent increments.
3. The number of events in any interval of length t is Poisson distributed with mean λt. That is, $\forall s, t \geq 0$,

$$\mathbb{P}\{N(t+s) - N(s) = n\} = \frac{e^{-\lambda t}(\lambda t)^n}{n!}, \quad n = 0, 1, \ldots$$

Poisson Process

Definition 2 of the Poisson Process

A Poisson process having rate λ is a counting process such that

1. $N(0) = 0$.
2. The process has stationary and independent increments.
3. $\mathbb{P}\{N(\delta) = 1\} = \lambda\delta + o(\delta)$.
4. $\mathbb{P}\{N(\delta) \geq 2\} = o(\delta)$.

Remark: The function f is said to be $o(h)$ if $\lim_{h \to 0} \frac{f(h)}{h} = 0$.

Exercise

Show equivalence between Definitions 1 and 2 of the Poisson process.

Example: Immigration

- Suppose that people immigrate into a territory at a Poisson rate $\lambda = 1$ per day.
- What is the expected time until the 10^{th} immigrant arrives?
- What is the probability that the elapsed time between the 10^{th} and the 11^{th} immigrants exceeds 2 days?

Further Properties of Poisson Process

Decomposition of a Poisson Process

- Consider a Poisson process $\{N_t, t \geq 0\}$ having rate $\lambda > 0$.
- Suppose that each time an event occurs it is randomly classified as either a type I or a type II.
- Suppose further that each event is classified as a type I event with probability p or a type II event with probability $1 - p$, independently of all other events.

Decomposition of a Poisson Process

Theorem

Let N_t^1 and N_t^2 denote respectively the number of type I and II events occurring in $[0, t]$. Then, $\{N_t^1, t \geq 0\}$ and $\{N_t^2, t \geq 0\}$ are both Poisson processes having respective rates λp and $\lambda(1-p)$. They are independent.

Superposition of Poisson Processes

Theorem

Let $\{N_t^1, t \geq 0\}$ and $\{N_t^2, t \geq 0\}$ be two independent Poisson processes having intensities λ_1 and λ_2, respectively. Then, $\{N_t^1 + N_t^2, t \geq 0\}$ is a Poisson process with intensity $\lambda_1 + \lambda_2$.

Example: Immigration Again

- If immigrants to area A arrive at a Poisson rate of 10 per week, and if each immigrant is of English descent with probability $1/12$, then what is the probability that no people of English descent will immigrate to area A in one month?

Example: Selling an Apartment

- Suppose that you want to sell your apartment. Potential buyers arrive according to a Poisson process with rate λ. Assume that each buyer offers you a random price following exponential distribution with rate μ. Once the offer is presented to you, you must either accept it or reject it and wait for the next offer.
- Suppose that you employ the policy of accepting the first offer that is greater than some pre-specified price y. Meanwhile, assume that a cost of rate c will be incurred per unit time until the apartment is sold.
- If your objective is to maximize the expected return, the difference between the amount received and the total cost incurred, what is the best price you should set?

Conditional Distribution of Arrival Times

Suppose that we are told that exactly one event of a Poisson process has taken place by time t, and we are asked to determine the distribution of the time at which the event occurred.

$$P(S_1 < s \mid N_t = 1) = \frac{s}{t}$$

Conditional Distribution of Arrival Times

- In other words, the time of the event should be uniformly distributed over $[0, t]$.
- This result may be generalized to the joint distribution of multiple arrivals.
    - Order statistics

Order Statistics

Let $Y_1, Y_2, \ldots, Y_n$ be n random variables. We say that $Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)}$ are the corresponding order statistics if $Y_{(k)}$ is the kth smallest value among $Y_1, Y_2, \ldots, Y_n$. For instance,

$$(Y_1, Y_2, Y_3) = (4, 5, 1)$$

The corresponding order statistics are

$$(Y_{(1)}, Y_{(2)}, Y_{(3)}) = (1, 4, 5).$$

Order Statistics

- If the $Y_i, i = 1, \ldots, n$, are independent identically distributed continuous random variables with PDF f, then the joint density of the order statistics $Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)}$ is given by

$$g(y_1, y_2, \ldots, y_n) = n! \prod_{i=1}^{n} f(y_i),$$

for $y_1 < y_2 < \cdots < y_n$.

Order Statistics

- In particular, if the $Y_i, i = 1, \ldots, n$, are uniformly distributed over $[0, t]$, then

$$g(y_1, y_2, \ldots, y_n) = \frac{n!}{t^n},$$

for $0 < y_1 < y_2 < \cdots < y_n < t$.

Conditional Distribution of Arrival Times

Theorem

Given that $N_t = n$, the n arrival times $S_1, S_2, \ldots, S_n$ have the same distribution as the order statistics corresponding to n independent random variables uniformly distributed on the interval $(0, t)$.

Example: Insurance Claims

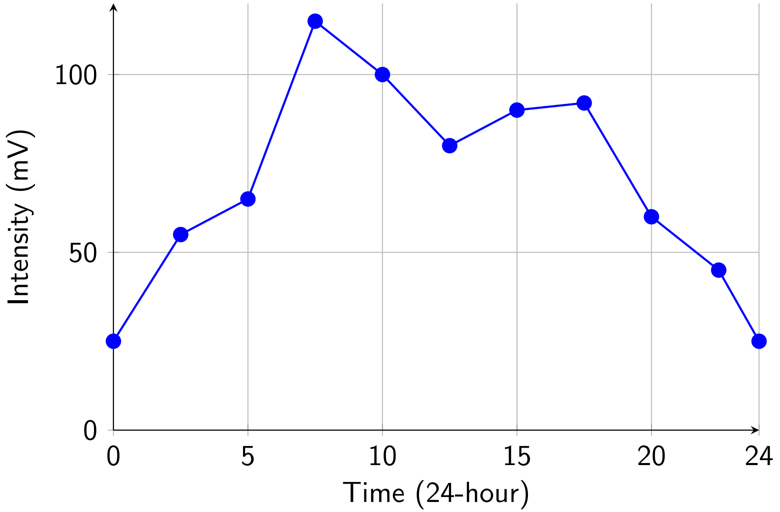
- Insurance claims are made at times distributed according to a Poisson process with rate λ; each claim amount is the same as $\$a$.
- Compute the discounted value of all claims made up to time t, where the discount factor is $\beta < 1$?.

Time-Inhomogeneous Poisson Process

Time Homogeneity

- The arrival intensity of a regular Poisson process is time homogenous: $N_{t+s} - N_t$ is a Poisson random variable with mean λs for any time t.
- But, in engineering practices, we do need non-homogenous Poisson processes to model non-stationary arrivals.

Arrival Intensity To A Call Center



Non-homogeneous Poisson Processes

- We may introduce a time-dependent intensity function $\lambda_t, t \geq 0$, to indicate the arrival rates at each moment t.
- For a non-homogeneous Poisson process N_t, its increment $N_{t+s} - N_t$ should be a Poisson random variable with mean

$$\int_t^{t+s} \lambda_u du.$$

Example: Siegbert's Hot Dog Stand

- Siegbert runs a hot dog stand that opens at 8am and closes at 5pm. He observes the following pattern about customer arrivals.
  - 8am to 11am: on average, a steadily increasing rate that starts with an initial rate of 5 customers per hour at 8am and reaches a maximum of 20 customers per hour at 11am
  - 11am to 1pm: remaining at 20 per hour
  - 1pm to 5pm: arrival rates drop steadily from 20 customers per hour to 12 customers per hour.
- What is the probability that no customers arrive between 8:30am to 9:30am?
- What is the expected number of arrivals in this period?

Generating Nonhomogeneous Poisson Processes

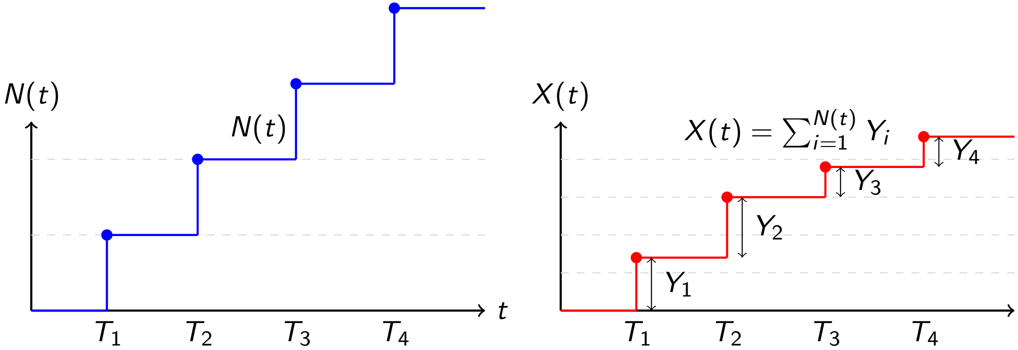
Idea: Generate events from a homogeneous Poisson process with rate $\lambda^* \geq \max_t \lambda(t)$, then accept each event with probability $\lambda(t)/\lambda^*$.

Algorithm:

1. Choose $\lambda^* \geq \sup_{t \geq 0} \lambda(t)$
2. Generate event times $T_1, T_2, \ldots$ from a homogeneous Poisson process with rate λ^*
3. For each event time T_i:
    - Generate $U_i \sim \text{Uniform}(0, 1)$
    - If $U_i \leq \lambda(T_i)/\lambda^*$, accept T_i
    - Otherwise, reject T_i
4. The accepted times form the NHPP

Compound Poisson Processes

An Illustration Figure



(A) Poisson Process **(B) Compound Poisson Process**

Example: Buses and Fans

- Suppose that buses arrive at a sporting event in accordance with a Poisson process, and suppose that the numbers of fans in each bus are i.i.d.

Example: Buses and Fans

Then $\{X_t, t \geq 0\}$ is a compound Poisson process, where

$$X_t = \sum_{i=1}^{N_t} Y_i, \quad t \geq 0$$

denotes the number of fans who have arrived by t, and Y_i represents the number of fans in the ith bus.

Mean and Variance

- Mean:
$$E[X_t] = \lambda t \cdot E[Y_1]$$
- Variance:
$$Var[X_t] = \lambda t \cdot E[Y_1^2].$$

Compound Poisson Process as a Superposition of Poisson Processes

- There is a very nice presentation of the compound Poisson process when the set of possible values of the Y is finite. Let

$$P(Y = \alpha_t) = p_t, \quad \sum_t p_t = 1.$$

- Let us say that the event is a type j event whenever the value of this arrival is $Y = \alpha_j$.
- Then, N_t^j, the number of type j events by time t, follows a Poisson process with rate λp_j.

Compound Poisson Process as a Superposition of Poisson Processes

Furthermore,

$$X_t = \sum \alpha_j N_{tj}.$$

Example: Jump Model for Security Price

- Let $\{S_t, t \geq 0\}$ be the price process of a financial security. A popular model supposes that the price remains unchanged until a "shock" occurs, at which time the price is multiplied by a random factor. That is,

$$S_t = S_0 \prod_{i=0}^{N_t} X_i,$$

where $\prod_{i=0}^{N_t} X_i = 1$ is equal to 1 when $N_t = 0$.

Example: Jump Model for Security Price

Suppose that $\{X_i\}_{i\in\{1,2,\cdots\}}$ are independent exponential random variables with rate μ; $\{N_t, t \geq 0\}$ is a Poisson process with rate λ and independent of $\{X_i\}_{i\in\{1,2,\cdots\}}$.

Find $E[S_t]$ and $E[S_t^2]$.

Mixed Poisson Processes

Definition

Let $\{N_t, t \geq 0\}$ be a counting process constructed as follows:

There is a positive random variable L such that, conditional on $L = \lambda$, the counting process is a Poisson process with rate λ.

Such a counting process is called a conditional or mixed Poisson process.

Distribution of Increments of Mixed Poisson Process

The distribution of the increment:

$$\begin{aligned} P(N_{t+s} - N_t = n) &= \int_0^{+\infty} P(N_{t+s} - N_t = n \mid L = \lambda) g(\lambda) d\lambda \\ &= \int_0^{+\infty} e^{-\lambda t} \frac{(\lambda t)^n}{n!} g(\lambda) d\lambda. \end{aligned}$$

Conditional Distribution of L

$$
\begin{aligned}
P(L \leq x \mid N_t = n) &= \frac{P(L \leq x, N_t = n)}{P(N_t = n)} \\
&= \frac{\int_0^{\infty} e^{-\lambda t}(\lambda t)^n g(\lambda) d\lambda}{\int_0^{x} e^{-\lambda t}(\lambda t)^n g(\lambda) d\lambda}
\end{aligned}
$$

Conditional Distribution of L

The conditional density function of L is thus given by

$$f_{L|N_t=n}(\lambda) = \frac{e^{-\lambda t}(\lambda t)^n g(\lambda)}{\int_0^\infty e^{-\lambda t}(\lambda t)^n g(\lambda) d\lambda}.$$

Example: Mixed Poisson Process with Gamma Mixing Distribution

- Consider an insurance company that sells policies to a heterogeneous population of drivers.
- Some drivers are cautious and have a low accident rate, while others are riskier and have a higher accident rate.
- If we knew each driver's type, we could model their accidents as a simple Poisson process.
- However, the insurance company typically does not observe each driver's risk type directly. This is **unobserved heterogeneity**.

Example: Mixed Poisson Process with Gamma Mixing Distribution

Let the mixing distribution L (the random accident rate) follow a **Gamma distribution** with shape parameter $\alpha > 0$ and rate parameter $\beta > 0$:

$$g(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \quad \lambda > 0$$

where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$ is the Gamma function.

Interpretation of parameters:

- α (shape) controls the heterogeneity (larger α means drivers are more similar)
- β (rate) scales the average rate: $\mathbb{E}[L] = \alpha/\beta$

Example: Mixed Poisson Process with Gamma Mixing Distribution

For a randomly selected driver, what is the probability of observing exactly n accidents in a time interval of length t?

- $P(N_t = n) = \binom{n+\alpha-1}{n} \left(\frac{\beta}{t+\beta}\right)^{\alpha} \left(\frac{t}{t+\beta}\right)^{n}, \quad n = 0, 1, 2, \ldots$
- This is the **negative binomial distribution**, one of the most common distributions for modeling overdispersed count data.

Overdispersion

For a standard Poisson process, the mean and variance are equal:

$$\mathbb{E}[N_t] = \mathrm{Var}(N_t) = \lambda t.$$

For our mixed Poisson process:

$$\mathbb{E}[N_t] = \alpha \cdot \frac{t}{\beta}, \quad \mathrm{Var}(N_t) = \alpha \cdot \frac{t}{\beta}\left(1 + \frac{t}{\beta}\right) = \mathbb{E}[N_t]\left(1 + \frac{t}{\beta}\right).$$

Overdispersion

- $\text{Var}(N_t) > \mathbb{E}[N_t]$, the process exhibits **overdispersion**.
- A hallmark of real-world count data: accidents, insurance claims, customer arrivals, and disease incidents typically show more variability than a simple Poisson model can capture.
- The mixed Poisson framework naturally generates this overdispersion by allowing rates to vary across individuals.

Example: Mixed Poisson Process with Gamma Mixing Distribution

Suppose we observe a specific driver for time t and see n accidents. What have we learned about their accident rate λ?

- $L \mid N_t = n \sim \text{Gamma}(n + \alpha,\, t + \beta)$

Takeaway

- **Before observing any accidents (prior)**: $\lambda \sim \text{Gamma}(\alpha, \beta)$
- **After observing *n* accidents in time *t* (posterior)**: $\lambda \sim \text{Gamma}(n + \alpha, t + \beta)$
- The shape parameter increases by the number of accidents observed (n)
- The rate parameter increases by the observation time (t)

Why This Framework is Powerful

1. **Handles heterogeneity naturally**: Different individuals can have different underlying rates without requiring us to observe them directly
2. **Generates overdispersion**: Matches the empirical fact that real count data is often more variable than Poisson
3. **Bayesian interpretation**: Provides a coherent framework for learning about individual rates as more data arrives
4. **Widely applicable**: Used in insurance (claim frequencies), healthcare (hospital visits), marketing (purchases), and many other fields